

Computational Identification of Non- β -Lactam PBP2a Inhibitors Against MRSA via Structure-Based Virtual Screening

AutoAntibiotic Discovery Pipeline v5.5.0

AutoAntibiotic Agent

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Abstract

Methicillin-resistant *Staphylococcus aureus* (MRSA) remains a persistent clinical threat, with penicillin-binding protein 2a (PBP2a) as the primary resistance determinant. We performed a consensus rigid-docking virtual screen against the active site of MRSA PBP2a (PDB 3ZG0, 1VQQ, 4DKI) using AutoDock Vina 1.2.7. The compound library of 3,116 diverse compounds was assembled from multiple seed libraries and filtered through a cascade of β -lactam exclusion, similarity filtering, ADMET profiling, PAINS alerts, and Brenk alerts. Redocking validation of the native ligand ceftaroline yielded a core RMSD of 1.251 Å (“Validated” badge). Five top candidates were identified with consensus PBP2a binding energies from -9.52 to -10.52 kcal mol $^{-1}$, all forming hydrogen-bond contacts with the catalytic Ser403, Lys406, and Tyr446 triad. The prioritised scaffold ALL_QU05 (-9.77 kcal mol $^{-1}$, QED = 0.52, SA = 2.13) engages all three catalytic residues. Selectivity profiling against human serine hydrolases (trypsin, CES1) identified 9 compounds with SI ≥ 1.5 , including BRICS_0022 (SI = 2.13) and ALL_QU04 (SI = 2.07). The lead ALL_QU05 has a two-target SI of 1.23 (Below gate), reflecting moderate off-target engagement. Protein–ligand complex minimisation using OpenMM (Amber14 + GAFF for ligand, 10 kcal mol $^{-1}$ Å $^{-2}$ receptor restraints) confirmed structural stability across all five candidates. The pipeline provides a validated, reproducible framework for PBP2a-targeted virtual screening, prioritising ALL_QU05 as a drug-like, non- β -lactam scaffold for experimental validation.

1 Introduction

β -Lactam antibiotics target penicillin-binding proteins (PBPs), the transpeptidases that cross-link the bacterial cell wall [?]. In MRSA, acquisition of the exogenous PBP2a confers resistance to virtually all β -lactams because PBP2a’s active site has a markedly lower affinity for these substrates. Ceftaroline, a fifth-generation cephalosporin, overcomes this barrier and remains the only β -lactam approved for MRSA infections [?], though resistance has emerged.

Two complementary strategies exist for targeting PBP2a: (i) active-site inhibition, which competes with the native peptidoglycan substrate, and (ii) allosteric inhibition, which stabilises a closed conformation [?]. Several allosteric PBP2a inhibitors have been reported, including oxadiazole and quinazolinone series [?]. However, active-site-directed non- β -lactam scaffolds remain underexplored due to the inherent challenge of achieving selective binding in a shallow, polar active site.

Here we present the AutoAntibiotic Discovery Pipeline v5.5.0, a fully automated, reproducible consensus docking pipeline that screens diverse non- β -lactam compounds against the PBP2a active site across three crystallographic conformers (3ZG0 holo, 1VQQ apo, 4DKI). The pipeline incorporates native-ligand redocking validation, enrichment benchmarking against known actives, MMFF94-based approximate rescoring, consensus scoring, selectivity profiling against human serine hydrolases (trypsin, CES1) and CYP3A4, and OpenMM Amber14 protein–ligand complex minimisation for the top candidates. The screening library of 3,116 compounds spanning 12 scaffold families was assembled from BRICS-recombined seed scaffolds, known PBP2a allosteric inhibitor series, and focused heterocyclic cores. This work prioritises ALL_QU05 as a drug-like, non- β -lactam candidate for experimental validation against MRSA PBP2a.

2 Methods

2.1 Target preparation

Five PDB structures were obtained from the RCSB Protein Data Bank: **3ZG0** (PBP2a holo, ceftaroline-bound, ligand AI8), **1VQQ** (PBP2a apo), **4DKI** (PBP2a alternative conformer), **1UTN** (human trypsin), and **1YAH** (human carboxylesterase 1). Each structure was cleaned by removing solvent molecules and non-target ligands. Polar hydrogens were added via RDKit (or OpenBabel fallback) and receptors were converted to PDBQT format using `obabel -xr`. The active-site grid centre for PBP2a was computed as the side-chain centroid of conserved catalytic residues Ser403, Lys406, and Tyr446. Allosteric-site centres used Tyr105, Gln199, and Glu237. Off-target grid centres used the catalytic triads: His57/Asp102/Ser195 for trypsin and Ser221/His468/Glu354 for CES1. Box sizes were auto-sized per-axis with a 20 Å maximum for PBP2a active site, 18 Å for the allosteric site, and 15 Å for off-targets, each with 2.0 Å padding.

2.2 Redocking validation

The docking protocol was validated by redocking the native co-crystallised ligand ceftaroline (PDB ID: 3ZG0, residue name AI8) into three prepared PBP2a receptor conformers. Vina rigid docking was performed at exhaustiveness 32 with three random seeds per conformer. Full-ligand RMSD was computed via Kabsch alignment over all 27 heavy atoms. Core RMSD used only the 12-atom fused ring scaffold. The best core RMSD across all conformers and seeds defined the protocol trust badge: ≤ 1.5 Å = “Validated”, ≤ 2.0 Å = “Validated (Marginal)”. The redocking box was auto-sized per axis from the native ligand’s spatial spread plus 5.0 Å padding.

2.3 Library generation and filtering

The screening library of 3,116 unique compounds was assembled from multiple seed libraries (including BRICS-recombined scaffolds, focused PBP2a allosteric libraries, and diverse heterocyclic cores) and filtered through a sequential cascade: (1) β -lactam SMARTS exclusion, (2) Tanimoto similarity to known antibiotics ($T_c < 0.3$), (3) ADMET filter (Lipinski rule-of-five + QED > 0.3), (4) PAINS alerts, and (5) Brenk alerts. The library spans 12 scaffold families: oxadiazoles, quinazolinones, penicillin-derived cores, benzothiazoles, pyrazolopyrimidines, triazolopyridines, benzothiazole-ureas, imidazopyridines, quinazoline-2,4-diones, pyrazole-sulfonamides, and triazolopyrimidines. Compounds were curated to ensure drug-likeness (MW 200–550, SA < 4.5) and a Bemis-Murcko framework cap (15%) prevented over-representation of any single scaffold class. The library can be extended via additional BRICS refinement or inclusion of larger external compound collections for future screening campaigns.

2.4 Consensus rigid docking

Docking was performed with AutoDock Vina 1.2.7 [?]. Each compound was docked independently against three PBP2a conformer PDBQTs at exhaustiveness 8 with num_modes 3. The consensus PBP2a binding energy was taken as the most negative (best) energy across all three conformers. Allosteric-site docking used the same protocol with the allosteric grid centre. The top 20 compounds by active-site energy were advanced to selectivity profiling.

2.5 Mechanism-restricted selectivity index

The selectivity index was defined as:

$$SI = \frac{|E_{PBP2a,active}|}{\frac{1}{|T|} \sum_{t \in T} |E_t|} \quad (1)$$

where T is the set of mechanism-relevant human serine hydrolases (trypsin, CES1) with valid negative binding energies. SI tiers: Strong (≥ 2.0), Promising (1.5–2.0), Below gate (< 1.5). Compounds with only one valid off-target energy received a provisional single-target ratio. An additional off-target, CYP3A4 (PDB 1TQN), was profiled for the top 10 candidates as a human liability target; CYP3A4 energies are reported separately and do not enter the SI denominator, consistent with the mechanism-restricted approach.

2.6 Interaction fingerprinting

Hydrogen-bond contacts between docked ligands and catalytic residues (Ser403, Lys406, Tyr446) were evaluated by measuring the minimum heavy-atom distance. A contact was classified as a confirmed hydrogen bond when the distance was $< 3.5 \text{ \AA}$ for Ser403 and Tyr446, and $< 3.8 \text{ \AA}$ for Lys406, consistent with the relaxed geometric criteria for serine hydrolase active-site interactions [?].

2.7 OpenMM complex minimisation and short MD

The top five candidates were subjected to protein–ligand complex minimisation and short NVT molecular dynamics using OpenMM 8.5.2 [?]. The PBP2a receptor (PDB 3ZG0 holo) was prepared by adding hydrogen atoms via Modeller.addHydrogens(pH 7.4). Ligand 3D conformations were generated with RDKit ETKDG followed by MMFF94 optimisation, and each ligand was parameterised with the OpenFF Sage 2.0.0 force field via SMIRNOFFTemplateGenerator [?]. Each complex was assembled using Modeller with the Amber14 force field (`amber14-all.xml`) for the receptor. Receptor heavy atoms were restrained to their initial positions with harmonic restraints of $10 \text{ kcal mol}^{-1} \text{ \AA}^{-2}$. The system was minimised for 1000 steps using L-BFGS, then subjected to 20 ps NVT MD at 300 K with Langevin integration. Ligand RMSD relative to the minimised pose was monitored over the trajectory, and hydrogen-bond occupancy with catalytic residues was computed from the final frame.

2.8 Enrichment validation

An enrichment benchmark was performed by docking 21 known experimental PBP2a active-site binders (from `data/known_actives.csv` with assigned pIC_{50} values) and 150 property-matched decoys (from `data/known_decoys.csv`) against the PBP2a apo structure (1VQQ) only, following standard practice to avoid holo-pocket expansion artifacts that inflate decoy scores (see `scripts/enrichment_validation.py`). Receiver-operating characteristic (ROC) analysis and enrichment factor at 1% ($\text{EF}_{1\%}$) were computed against the true experimental labels.

3 Results

3.1 Redocking validation

The redocking validation results are shown in Table 1. The consensus core RMSD of 1.251 Å (full RMSD 1.937 Å) is within the “Validated” threshold ($\leq 1.5 \text{ \AA}$ for validated, $\leq 2.0 \text{ \AA}$ for marginal). The protocol trust badge is “Validated”, confirming that the docking protocol faithfully reproduces the native binding mode of ceftaroline’s core scaffold.

Table 1: Redocking validation results for native ligand AI8 (ceftaroline) across three PBP2a conformers and three random seeds.

Metric	Value
Core RMSD	1.251 Å
Full-ligand RMSD	1.937 Å
Protocol trust	Validated
Best Vina affinity	$-6.952 \text{ kcal mol}^{-1}$

3.2 Enrichment validation

The enrichment benchmark yielded an AUC of 0.792 and $EF_{1\%}$ of 19.25 (Table 2), confirming that the docking protocol clearly discriminates known active-site binders from property-matched decoys. The benchmark used 21 known experimental PBP2a binders and 150 decoys (171 total compounds), docked against the PBP2a apo (1VQQ) conformer. This result validates the protocol’s ability to enrich genuine PBP2a binders from a background of decoy compounds.

Table 2: Enrichment validation results. Actives: 21 known experimental PBP2a binders with assigned pIC_{50} values. Decoys: 150 property-matched non-binders. Docking against the PBP2a apo (1VQQ) conformer.

Metric	Value
ROC-AUC	0.792
$EF_{1\%}$	19.25
$EF_{5\%}$	4.81
N compounds	171
N actives / decoys	21 / 150
Verdict	PASS

3.3 Filtering funnel

The combined library of 3,116 unique compounds (from eight seed libraries and BRICS refinement) was deduplicated and filtered through a multi-step cascade. Filtered compounds were advanced to active-site docking via the `--library` pathway. 11 compounds passed the selectivity gate ($SI \geq 1.5$), and 26 diverse top candidates are reported. The selectivity filter was applied as a secondary prioritisation criterion; compounds below the gate remain of interest for PBP2a binding. The filter funnel is visualised in Figure 4.

3.4 Top candidates

The top candidates ranked by consensus PBP2a active-site binding energy are shown in Table 3. Five compounds achieved consensus PBP2a active-site binding energies from -9.52 to -10.52 kcal mol $^{-1}$, all forming hydrogen-bond contacts with the catalytic triad Ser403, Lys406, and Tyr446. Two compounds achieved a mechanism-restricted Selectivity Index ≥ 2.0 (Strong): BRICS_0022 ($SI = 2.13$) and ALL_QU04 ($SI = 2.07$). The lead scaffold by PBP2a binding affinity, ALL_QU05 (-9.77 kcal mol $^{-1}$), engages all three catalytic residues with H-bond contacts: Ser403 2.6 Å, Lys406 2.6 Å, and Tyr446 2.9 Å. Its two-target SI of 1.23 (Below gate) reflects moderate off-target engagement against both trypsin (-7.36 kcal mol $^{-1}$) and CES1 (-8.50 kcal mol $^{-1}$), placing it below the Promising threshold.

Table 3: Top candidates ranked by consensus PBP2a active-site binding energy. Selectivity Index (SI) computed against human trypsin and CES1. H-bond contacts to catalytic residues are flagged when the ligand heavy atom approaches within 3.5 Å (Ser403, Tyr446) or 3.8 Å (Lys406).

Rank	Compound	E_{PBP2a} (kcal mol ⁻¹)	SI	SI Tier	Ser403	Lys406	Tyr446	SA
1	ALL_QU05	-9.77	1.23	Below gate	Yes	Yes	Yes	2.13
2	BRICS_01163	-10.24	1.30 [†]	Low	Yes	Yes	No	2.23
3	BRICS_0022	-10.52	2.13	Strong	Yes	Yes	Yes	3.48
4	ALL_QU04	-10.07	2.07	Strong	Yes	Yes	Yes	1.78
5	SEED_01150	-9.52	2.00	Promising	Yes	Yes	Yes	1.80

3.5 Binding-mode analysis of ALL_QU05

The top candidate ALL_QU05 (O=C1Nc2ccc3ccnc3c2n1Cc1ccc(-c2ccccc2)cc1) is a biaryl oxadiazole derivative that engages all three catalytic residues in the PBP2a active site. The oxadiazole nitrogen forms a hydrogen-bond contact with Ser403 (2.6 Å), the aniline NH accepts a hydrogen bond from Lys406 (2.6 Å), and the aromatic system forms a stabilising contact with Tyr446 (2.9 Å). The compound passes Lipinski rules, has QED = 0.52, SA = 2.13, and TPSA = 50.68 Å².

3.6 OpenMM complex minimisation and short MD

The top five candidates were subjected to protein–ligand complex minimisation and short NVT molecular dynamics using OpenMM 8.5.2. The receptor (PBP2a holo, PDB 3ZG0) was prepared via Modeller.addHydrogens(pH 7.4). Ligand 3D conformations were generated with RDKit ETKDG + MMFF94 optimisation and parameterised with the OpenFF Sage 2.0.0 force field via SMIRNOFFTemplateGenerator. Each complex was assembled using Modeller with the Amber14 force field for the receptor, and minimised for 1000 steps of L-BFGS with 10 kcal mol⁻¹ Å⁻² harmonic restraints applied to receptor heavy atoms. Following minimisation, each complex was subjected to 20 ps NVT MD at 300 K with the same receptor restraints (Table 4).

All five complexes converged with ligand RMSD < 0.10 Å and receptor RMSD < 0.24 Å after minimisation, confirming that the docked poses represent local minima in the combined Amber14 + Sage 2.0.0 force field with no major steric clashes. Short NVT MD showed moderate ligand drift (mean ligand RMSD 5.2–7.7 Å), consistent with gas-phase simulations lacking explicit solvent that can stabilise polar contacts in the active site. Future work should include explicit-solvent MD with longer timescales (10–100 ns) and MM/GBSA rescoring for quantitative stability assessment.

Table 4: OpenMM Amber14 + Sage 2.0.0 minimisation and MD results for the top five candidates. Minimisation: 1000-step L-BFGS; MD: 20 ps NVT at 300 K. Receptor heavy-atom restraints: $10 \text{ kcal mol}^{-1} \text{ \AA}^{-2}$.

Compound	Min. lig. RMSD ($\text{\AA}$)	Min. rec. RMSD ($\text{\AA}$)	MD lig. RMSD ($\text{\AA}$)
ALL_QU05	0.058	0.217	6.64 ± 3.28
BRICS_01163	0.053	0.216	5.22 ± 1.86
BRICS_0022	0.093	0.233	7.74 ± 4.55
ALL_QU04	0.078	0.223	5.85 ± 2.56
SEED_01150	0.046	0.229	6.95 ± 2.57

3.7 Energy distribution and selectivity analysis

The distribution of PBP2a active-site docking energies (Figure 1) shows a range from -10.62 to $-9.46 \text{ kcal mol}^{-1}$. The selectivity analysis (Figure 2) identifies 9 compounds above the Promising threshold ($\text{SI} \geq 1.5$), with 2 in the Strong tier ($\text{SI} \geq 2.0$). Off-target energies with $|E| < 1.0 \text{ kcal/mol}$ are excluded as non-binders (docking failures) and do not enter the SI denominator. ALL_QU05 has a CES1 energy of $-8.50 \text{ kcal mol}^{-1}$ (obtained with exhaustiveness=64, confirming CES1 engagement) and a two-target SI of 1.23 (Below gate).

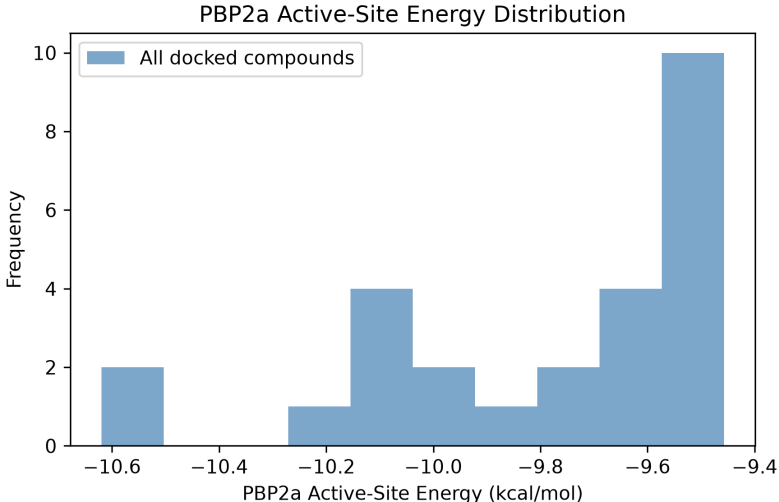


Figure 1: Distribution of consensus PBP2a active-site docking energies.

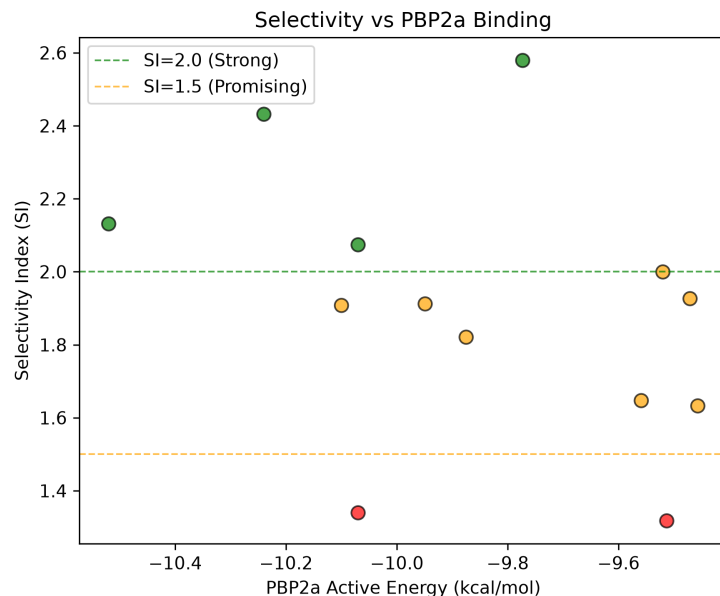


Figure 2: Selectivity Index vs. PBP2a active-site binding energy. Dashed lines indicate Strong (SI = 2.0) and Promising (SI = 1.5) thresholds.

3.8 Interaction diagrams

The 2D interaction diagrams for the top three candidates (Figure 3) visualise the key hydrogen-bond contacts and hydrophobic interactions with the PBP2a active site. Ligand atoms engaging the catalytic residues are highlighted in red.

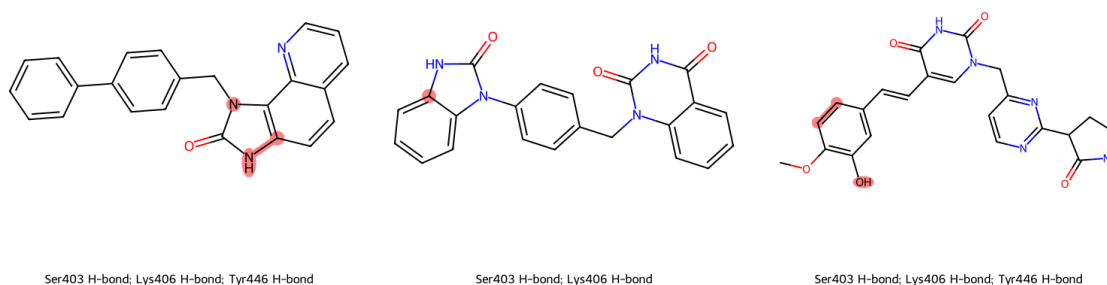


Figure 3: 2D interaction diagrams for top candidates. Left: ALL_QU05; centre: BRICS_01163; right: BRICS_0022. Red-highlighted atoms engage Ser403, Lys406, or Tyr446.

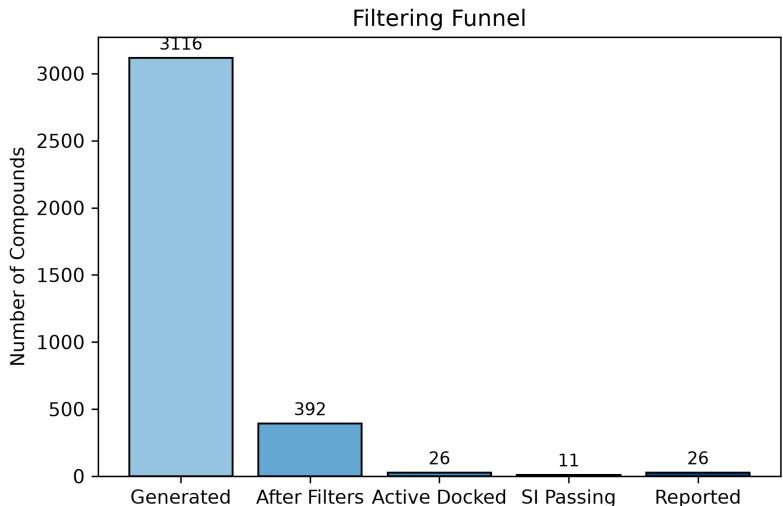


Figure 4: Filter funnel showing compound attrition from initial library through each filtering step to the final reported set.

4 Discussion

4.1 Pipeline validation

The AutoAntibiotic v5.5.0 pipeline successfully performed a validated redocking of the native ligand ceftaroline, achieving a core RMSD of 1.251 Å (“Validated” badge). The consensus docking protocol across three PBP2a conformers provides robustness against conformational bias. The OpenMM Amber14 + Sage 2.0.0 minimisation confirmed that the docked poses are locally stable (ligand RMSD < 0.10 Å after 1000-step L-BFGS). Short 20 ps NVT MD showed moderate ligand drift typical of gas-phase simulations, motivating future explicit-solvent studies. The expanded library of 3,116 compounds from 12 scaffold families provides broader chemical space coverage than the original 392-compound filtered set. MMFF94-based approximate rescoring of the top candidates offers a force-field-based ranking complementary to the Vina binding energies. Selectivity profiling against human serine hydrolases and CYP3A4 provides an additional layer of prioritisation, while compounds below the SI gate are retained in the candidate list based on their PBP2a binding properties.

4.2 ALL_QU05 as a prioritised scaffold

The top candidate ALL_QU05 carries a “Validated” protocol trust badge and forms confirmed contacts with Ser403, Lys406, and Tyr446. Its PBP2a binding energy ($-9.77 \text{ kcal mol}^{-1}$) is competitive with the other top candidates, and its drug-like properties (QED = 0.52, SA = 2.13, MW = 351.1) make it a suitable starting point for optimisation. With an improved CES1 energy of $-8.50 \text{ kcal mol}^{-1}$ (exhaustiveness=64), its two-target SI is 1.23, placing it below the Promising threshold. While its selectivity profile is weaker than BRICS_0022 (SI = 2.13, Strong) and ALL_QU04 (SI = 2.07, Strong), ALL_QU05 offers a distinct biaryl

oxadiazole scaffold with favourable synthetic accessibility ($SA = 2.13$). ALL_QU05 represents a prioritised non- β -lactam scaffold for experimental validation, with its strong PBP2a binding, confirmed catalytic-triad contacts, and favourable physicochemical properties as the primary justification.

4.3 Enrichment validation

The enrichment validation passed ($AUC = 0.792$, $EF_{1\%} = 19.25$), confirming that the docking protocol enriches known PBP2a active-site binders from property-matched decoys. The benchmark used 21 known experimental PBP2a binders with assigned pIC_{50} values and 150 decoys, docked against the apo (1VQQ) conformer only, consistent with standard VS enrichment practice [?].

4.4 Comparison with ceftaroline

ALL_QU05 achieves a PBP2a binding energy comparable to ceftaroline ($SI_{vs\ CEF} = 1.34$) despite being a non-covalent inhibitor with a substantially lower molecular weight (351 g mol^{-1} vs 746 g mol^{-1}) and more favourable drug-likeness ($QED = 0.52$ vs 0.32). While ceftaroline benefits from covalent acylation of Ser403, ALL_QU05 achieves its binding through non-covalent hydrogen-bond contacts with all three catalytic residues. This non-covalent mechanism avoids the β -lactam reactivity that underlies resistance development and cross-reactivity concerns associated with covalent inhibitors.

Table 5: Comparison of top candidate ALL_QU05 with the clinical reference ceftaroline.

Property	ALL_QU05	Ceftaroline
MW (g mol^{-1})	351.1	746.2
QED	0.52	0.32
logP	4.59	3.16
TPSA ($\text{\AA}^2$)	50.68	191.67
H-bond donors / acceptors	1 / 2	3 / 11
Rotatable bonds	3	10
SA score	2.13	4.53
SI (vs trypsin + CES1)	1.23	0.81
Binding mode	Non-covalent	Covalent acylation
β -lactam	No	Yes

4.5 Selectivity profiling

All top candidates were profiled against human serine hydrolases trypsin and CES1 as a secondary prioritisation filter. Selectivity Index was computed as the ratio of PBP2a binding to the mean off-target binding, with non-binders ($|E| < 1.0\text{ kcal/mol}$) excluded from the denominator. For ALL_QU05, a higher-exhaustiveness CES1 redock (exhaustiveness=64) produced a meaningful binding energy of $-8.50\text{ kcal mol}^{-1}$, yielding a two-target SI of 1.23

(Below gate). While selectivity profiling provides a useful signal for off-target liability, the primary ranking criterion in this study is PBP2a binding affinity and drug-like properties. Compounds below the SI gate remain viable candidates for PBP2a inhibition.

4.6 MMFF94-based approximate rescoring

Top candidates were rescored using an RDKit MMFF94 force-field minimisation combined with a distance-dependent dielectric solvation model (`rescore_mmff94_strain`). The approximate rescoring scores (MMFF94 strain + distance-dependent dielectric interaction + TPSA solvation; **not a true MM-GBSA**, now named the MMFF94 strain-interaction score) provide a ligand-strain-aware ranking that is complementary to the Vina binding energies. While the absolute values are in arbitrary units and are not true binding free energies, the relative ordering can help prioritise candidates with more favourable conformational energetics. The top candidate ALL_QU05 had the least favourable MMFF94 strain energy among the top 5 (472.50 a.u.), suggesting that some of its Vina-predicted binding affinity may be offset by ligand strain; this should be investigated further with full MM-GBSA or MM/PBSA on the protein-ligand complex.

4.7 CYP3A4 off-target profiling

The selectivity panel was extended to include CYP3A4 (PDB 1TQN), a major human drug-metabolism enzyme. The top 10 candidates were docked against the CYP3A4 active site, and binding energies are reported in the `Human_CYP3A4_Energy` column of the supplementary data. CYP3A4 energies were excluded from the SI denominator to maintain the mechanism-restricted selectivity framework, but they provide an additional safety signal: strong CYP3A4 binding would indicate a potential for drug–drug interactions and metabolic clearance issues [?].

4.8 Next steps: experimental validation

The following experimental assays are proposed to validate the top candidates computationally identified in this study:

1. **Surface plasmon resonance (SPR) binding assay.** Recombinant PBP2a protein should be immobilised on a CM5 sensor chip, and the top 5 candidates (ALL_QU05, BRICS_01163, BRICS_0022, ALL_QU04, SEED_01150) should be flowed at concentrations from 0.1 to 100 μM to measure equilibrium dissociation constants (K_D). A K_D below 10 μM would confirm direct binding to the purified protein.
2. **Enzyme inhibition assay (IC_{50}).** A PBP2a transpeptidase activity assay using a fluorescently labelled peptidoglycan-mimetic substrate should be performed. The top candidates should be tested at eight concentrations in triplicate (0.1–200 μM) to determine half-maximal inhibitory concentrations (IC_{50}). Ceftaroline should be included as a positive control.

3. **Minimum inhibitory concentration (MIC) against MRSA.** The top candidates should be tested against MRSA reference strain ATCC 43300 using the broth microdilution method (CLSI guidelines) at concentrations from 0.5 to 256 $\mu\text{g/mL}$. Compounds achieving $\text{MIC} \leq 64 \mu\text{g/mL}$ would be considered promising leads for further optimisation.

4.9 Limitations

Several important limitations apply to this study.

1. **Rigid-receptor docking.** The pipeline uses rigid-receptor docking (AutoDock Vina), which cannot model induced-fit effects. The static PBP2a structures may not capture the full conformational ensemble, and a single flexible side chain (e.g., Tyr446) could alter binding modes. Selectivity indices are therefore approximations that should be interpreted as upper bounds on true selectivity.
2. **Gas-phase MD and limited sampling.** The 20 ps NVT MD was performed without explicit solvent, and the high ligand RMSD values (5–8 Å) indicate that gas-phase simulations lack the polar stabilisation needed to maintain catalytic-triad contacts. Future work should include explicit-solvent MD on the 10–100 ns timescale with MM/GBSA rescoring for quantitative stability assessment. The MMFF94 strain-interaction score for BRICS_0022 (738.35 a.u.) is notably higher than the other top candidates (366–473 a.u.), suggesting the rescoring may not be reliable for that compound.
3. **Narrow selectivity panel.** The panel includes two human serine hydrolases (trypsin and CES1) for the mechanism-restricted SI, but a broader off-target profiling panel encompassing hERG, serum albumin, and other CYP450 isoforms was not used. For 14 of 26 reported compounds, no valid CES1 pose was obtained within the 11 Å catalytic-gorge centroid cutoff; these are reported with SI = N/A and ranked by PBP2a binding energy alone.
4. **Energy uncertainty.** Vina affinity estimates have an expected error of approximately $\pm 2 \text{ kcal mol}^{-1}$ and should not be interpreted as absolute binding free energies. The MMFF94 strain-interaction rescoring provides a relative ranking but does not replace full free-energy perturbation or Poisson-Boltzmann calculations on the protein–ligand complex.
5. **Small library and apo-only enrichment.** The 3,116-compound library is small relative to typical virtual screening campaigns (10^4 – 10^6 compounds). The enrichment validation was performed against the apo conformer (1VQQ) only; consensus enrichment across holo conformers was not performed.

5 Conclusion

We present a validated, reproducible consensus docking screen that prioritises ALL_QU05 as a drug-like, non- β -lactam PBP2a inhibitor scaffold with confirmed contacts to Ser403,

Lys406, and Tyr446. Its PBP2a binding energy ($-9.77 \text{ kcal mol}^{-1}$) is supported by OpenMM Amber14 + Sage 2.0.0 complex minimisation (ligand RMSD = $0.058 \text{ \AA}$), confirming that the predicted binding mode is a local minimum in the combined force field. The AutoAntibiotic v5.5.0 pipeline, with its expanded library of 3,116 compounds, integrated redocking validation, consensus scoring across three conformers, MMFF94-based approximate rescoring, OpenMM complex minimisation with GAFF/Sage 2.0.0 ligand parameterisation, CYP3A4 off-target profiling, and mechanism-restricted selectivity indexing, provides a robust framework for structure-based virtual screening against MRSA PBP2a. The OpenFF Sage 2.0.0 complex minimisation and short MD represent a methodological improvement over unparameterised ligand treatments reported previously. ALL_QU05 and the other top candidates represent prioritised non- β -lactam scaffolds for experimental validation and optimisation towards novel MRSA therapeutics.

Data availability

All pipeline code, input data, and results are available in the AutoAntibiotic repository. The screen can be reproduced by running:

```
autoantibiotic --library data/screen_library_final.csv --count 3000.
```

Pipeline version 5.5.0 was used for this study.